

DANDI:001333: Parkinson's Electrophysiological Signal Dataset (PESD)

Biswas, A. (2025). *Parkinson's Electrophysiological Signal Dataset (PESD)* (Version 0.250327.2220) [Data set]. DANDI Archive. doi:10.48324/dandi.001333/0.250327.2220. BrainX Lab, Michigan Technological University. Companion paper: arXiv:2407.17756. Converted with `dandi_export/export_001333.py`.

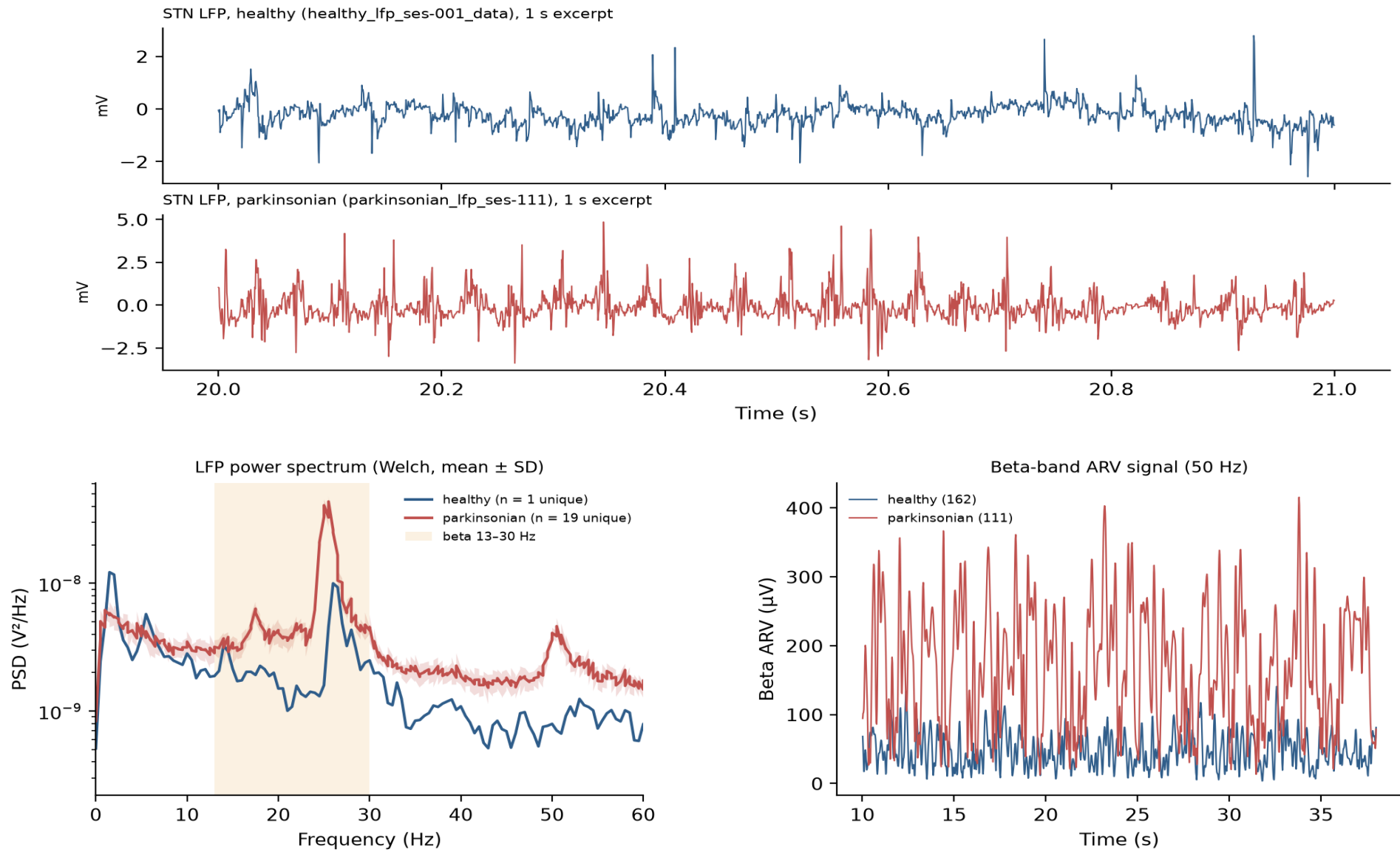
These are simulated signals, not recordings. All 52 files come from the computational cortico-basal-ganglia network model of Fleming et al. (2020, *Front. Neurosci.* 14:166, doi:10.3389/fnins.2020.00166), run in NEURON. The 'subjects' are simulation conditions; species 'Homo sapiens', age 0 and sex 'U' in the files are placeholders. **The 19 duplicate files:** the 10 healthy beta-ARV files are byte-identical, and the 10 healthy LFP files plus 'healthy-simulated-data' are byte-identical, so the healthy condition is a **single simulation run**. Only 33 of the 52 files are unique. Healthy-vs-parkinsonian statistics that count the healthy files as independent samples are pseudo-replicated.

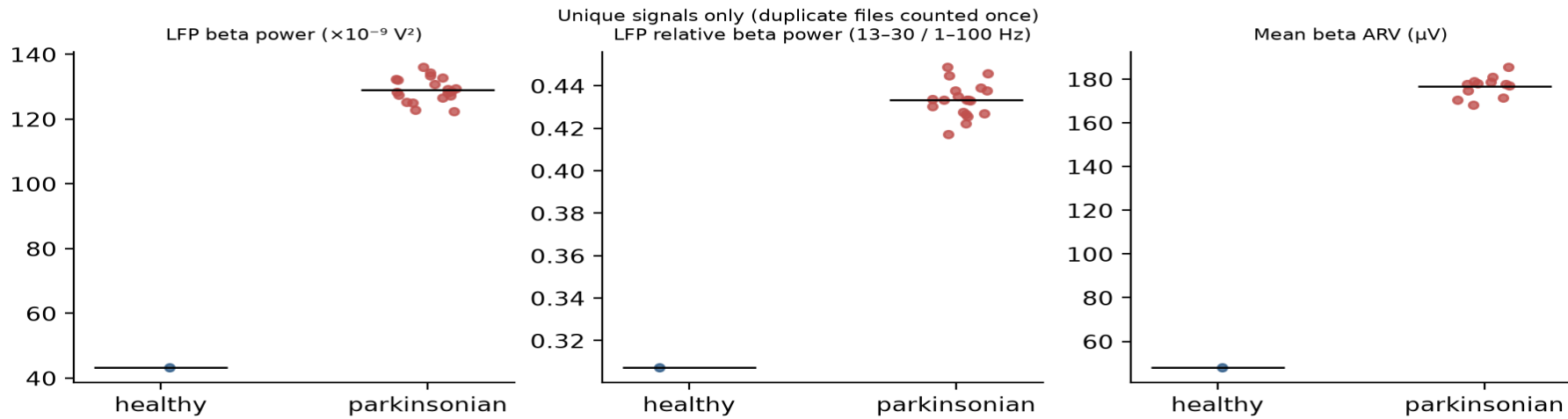
Group	Signal	Files	Unique signals
healthy	beta_arv	10	1
healthy	lfp	11	1
parkinsonian	beta_arv	12	12
parkinsonian	lfp	19	19

1. Parameters

Group	Parameter	Value
Source	Model	Cortico-basal-ganglia network (cortex, STN, GPe, GPi, thalamus), Fleming et al. 2020; simulated in NEURON
Source	Conditions	healthy vs parkinsonian network parameters (parkinsonian shows exaggerated 13–30 Hz beta oscillations in STN)
Source	'Electrodes'	12 virtual contacts (4 shanks × 3) on a 'NEURON_Simulator' virtual probe; location 'Simulated Cortico-basal-ganglia network of brain'
LFP	Signal	STN local field potential, 1 channel per file (V)
LFP	Sampling rate	2000 Hz
LFP	Duration	38.7 s (23 files), 32.75 s (7 files)
Beta ARV	Definition	average rectified value of the beta band: LFP band-pass filtered with a 4th-order Chebyshev filter, 8 Hz bandwidth centred on the LFP spectral peak, then fully rectified and averaged (from the dataset description)
Beta ARV	Sampling / span	50 Hz, t = 10.02–38.00 s (1400 samples)
Export	Power spectrum	Welch, 2 s Hann segments (0.5 Hz resolution), mean removed, 0–100 Hz kept
Export	Beta power	integral of PSD over 13–30 Hz; relative = beta / 1–100 Hz power; peak searched 5–45 Hz
Export	Duplicates	MD5 of each file's signal; duplicate_of names the first file with identical data

2. Figures





3. Group summary (unique signals only)

Group	Signal	n	Beta power mean (V^2)	Beta power SD	Relative beta mean	Relative beta SD	Peak freq (Hz)	Signal SD mean (V)	Mean value (V)
healthy	beta_arv	1						2.469e-05	4.814e-05
healthy	lfp	1	4.333e-08		0.307		26	5.429e-04	-2.306e-04
parkinsonian	beta_arv	12						8.975e-05	1.767e-04
parkinsonian	lfp	19	1.291e-07	3.81e-09	0.433	0.008	25, 25.5, 25.5	8.416e-04	-9.347e-05

Parkinsonian LFP beta power is 3.0 $\times$ the single healthy run's.

4. All files

Duplicate files are highlighted.

Sample	Group	Signal	Samples	fs (Hz)	Duration (s)	Unique	Duplicate of	SD (V)	Beta power (V ²)	Rel. beta	Peak (Hz)
healthy_beta_arv_ses-162	healthy	beta_arv	1400	50	28.00	yes		2.469e-05			
healthy_beta_arv_ses-218	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-279	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-345	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-423	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-452	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-516	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-54	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-646	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-78	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
parkinsonian_beta_arv_ses-111	parkinsonian	beta_arv	1400	50	28.00	yes		8.493e-05			
parkinsonian_beta_arv_ses-112	parkinsonian	beta_arv	1400	50	28.00	yes		9.117e-05			
parkinsonian_beta_arv_ses-120	parkinsonian	beta_arv	1400	50	28.00	yes		8.874e-05			
parkinsonian_beta_arv_ses-163	parkinsonian	beta_arv	1400	50	28.00	yes		9.226e-05			
parkinsonian_beta_arv_ses-164	parkinsonian	beta_arv	1400	50	28.00	yes		8.848e-05			
parkinsonian_beta_arv_ses-284	parkinsonian	beta_arv	1400	50	28.00	yes		9.005e-05			
parkinsonian_beta_arv_ses-289	parkinsonian	beta_arv	1400	50	28.00	yes		9.216e-05			
parkinsonian_beta_arv_ses-298	parkinsonian	beta_arv	1400	50	28.00	yes		8.858e-05			
parkinsonian_beta_arv_ses-339	parkinsonian	beta_arv	1400	50	28.00	yes		8.851e-05			
parkinsonian_beta_arv_ses-51	parkinsonian	beta_arv	1400	50	28.00	yes		8.864e-05			
parkinsonian_beta_arv_ses-60	parkinsonian	beta_arv	1400	50	28.00	yes		8.987e-05			
parkinsonian_beta_arv_ses-93	parkinsonian	beta_arv	1400	50	28.00	yes		9.362e-05			
healthy_lfp_ses-001_data	healthy	lfp	77400	2000	38.70	yes		5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-162	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-218	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-279	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-345	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-423	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-452	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-516	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-54	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-646	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-78	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
parkinsonian_lfp_ses-111	parkinsonian	lfp	77400	2000	38.70	yes		8.367e-04	1.249e-07	0.430	25.0
parkinsonian_lfp_ses-112	parkinsonian	lfp	77400	2000	38.70	yes		8.397e-04	1.276e-07	0.434	25.5
parkinsonian_lfp_ses-120	parkinsonian	lfp	77400	2000	38.70	yes		8.450e-04	1.284e-07	0.422	25.0
parkinsonian_lfp_ses-124	parkinsonian	lfp	65500	2000	32.75	yes		8.477e-04	1.293e-07	0.428	25.5
parkinsonian_lfp_ses-159	parkinsonian	lfp	65500	2000	32.75	yes		8.361e-04	1.224e-07	0.417	25.5
parkinsonian_lfp_ses-163	parkinsonian	lfp	77400	2000	38.70	yes		8.421e-04	1.308e-07	0.446	25.0

Sample	Group	Signal	Samples	fs (Hz)	Duration (s)	Unique	Duplicate of	SD (V)	Beta power (V ²)	Rel. beta	Peak (Hz)
parkinsonian_lfp_ses-164	parkinsonian	lfp	77400	2000	38.70	yes		8.315e-04	1.267e-07	0.438	25.0
parkinsonian_lfp_ses-219	parkinsonian	lfp	65500	2000	32.75	yes		8.521e-04	1.334e-07	0.426	25.5
parkinsonian_lfp_ses-245	parkinsonian	lfp	65500	2000	32.75	yes		8.491e-04	1.295e-07	0.427	25.5
parkinsonian_lfp_ses-284	parkinsonian	lfp	77400	2000	38.70	yes		8.426e-04	1.282e-07	0.433	25.0
parkinsonian_lfp_ses-289	parkinsonian	lfp	77400	2000	38.70	yes		8.392e-04	1.323e-07	0.445	25.5
parkinsonian_lfp_ses-298	parkinsonian	lfp	77400	2000	38.70	yes		8.432e-04	1.271e-07	0.434	25.0
parkinsonian_lfp_ses-318	parkinsonian	lfp	65500	2000	32.75	yes		8.412e-04	1.320e-07	0.433	25.5
parkinsonian_lfp_ses-339	parkinsonian	lfp	77400	2000	38.70	yes		8.464e-04	1.327e-07	0.435	25.5
parkinsonian_lfp_ses-51	parkinsonian	lfp	77400	2000	38.70	yes		8.322e-04	1.252e-07	0.433	25.5
parkinsonian_lfp_ses-60	parkinsonian	lfp	77400	2000	38.70	yes		8.376e-04	1.285e-07	0.438	25.0
parkinsonian_lfp_ses-66	parkinsonian	lfp	65500	2000	32.75	yes		8.453e-04	1.343e-07	0.439	25.5
parkinsonian_lfp_ses-69	parkinsonian	lfp	65500	2000	32.75	yes		8.336e-04	1.228e-07	0.427	25.5
parkinsonian_lfp_ses-93	parkinsonian	lfp	77400	2000	38.70	yes		8.487e-04	1.361e-07	0.449	25.5

5. Data files and columns

File	Rows	Columns
tables/sessions.csv	52	sample_id, file, group, signal, session, nwb_series, n_samples, fs_hz, t_start_s, t_end_s, duration_s, md5, duplicate_of, unique, mean_v, sd_v, rms_v, min_v, max_v, beta_power_v2, total_power_1_100_v2, relative_beta_power, peak_freq_hz, peak_psd_v2_per_hz
tables/psd.csv	6,000	sample_id, group, unique, freq_hz, psd_v2_per_hz (LFP files only, 0–100 Hz)
signals/<sample_id>.csv	one file per NWB file	time_s, voltage_v (the raw signal, SI units)

For modelling, filter `sessions.csv` to `unique == True`. Beta-ARV and LFP files sharing a session number within a group appear to be paired (the ARV derived from that LFP), but the files do not state this explicitly. Column descriptions: [dandi_export/README_001333.md](#).